

CUSTOMER CHURN ANALYSIS

Future Interns Data Science Internship - Task 2

Analysis using Python and Power BI

Project Report

Submitted by

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Abstract

This report describes the work I did for Task 2 of the Future Interns Data Science Internship, which was a customer churn analysis. I used the Telco Customer Churn dataset, which has 7,043 customer records and 21 columns. Using Python (Pandas and Matplotlib), I inspected the data, calculated the main churn KPIs and compared churn across contract type, internet service, customer tenure, monthly charges, payment method, Tech Support and Online Security. I then built a two-page interactive report in Power BI Desktop with KPI cards, slicers, comparison charts and a contract-by-internet-service churn-rate matrix.

The overall churn rate in the dataset is 26.54%. Churn is much higher among customers in their first six months (52.94%), customers who pay by electronic check (45.29%), month-to-month customers (42.71%) and fiber optic users (41.89%). Customers without Tech Support or Online Security also churn at around 42%. Based on these patterns, I have suggested some retention actions, such as stronger onboarding for new customers and encouraging longer contracts. Since the analysis is descriptive, these results show where churn is high but do not prove what causes it.

Table of Contents

1. Introduction	4
2. Objectives.....	4
3. Dataset Description	4
4. Tools and Technologies Used.....	5
5. Methodology	5
5.1 Data Loading and Inspection	5
5.2 Data Preparation.....	5
5.3 KPI Calculation	6
5.4 Segment Analysis.....	6
5.5 Dashboard Development	6
5.6 Business Interpretation	6
6. Overall Churn KPIs	6
7. Analysis and Key Findings	7
7.1 Contract Type	7
7.2 Internet Service.....	7
7.3 Customer Tenure	8
7.4 Payment Method.....	8
7.5 Tech Support and Online Security.....	8
7.6 Monthly Charges.....	8
7.7 Contract Type and Internet Service Combined	8
7.8 Summary of Findings.....	8
8. Power BI Dashboard	9
8.1 Page 1: Churn Dashboard.....	9
8.2 Page 2: Insights & Recommendations	10
9. Retention Recommendations	10
10. Challenges Faced.....	11
11. Learning Outcomes	11
12. Limitations and Future Scope	12
13. Conclusion	12
14. References.....	12
Appendix: Project Files.....	12

1. Introduction

Customer churn refers to customers who stop using a company's service. For a telecom company, every customer who leaves means a loss of monthly revenue, so it is important to know which types of customers are more likely to leave. If these customers can be identified early, the company can take steps to keep them.

For Task 2 of my Future Interns Data Science Internship, I carried out a customer churn analysis on the Telco Customer Churn dataset, which contains customer data from a telecom company. The aim was to find out which customer groups have higher churn and to present the results in a way that supports business decisions. I used Python to explore the data and calculate churn rates, and Power BI to build an interactive dashboard. The work produced for this task includes the Python analysis, a two-page Power BI dashboard and this report.

2. Objectives

The main objectives of this task were:

1. To measure the overall customer churn level using clear KPIs.
2. To explore how churn varies across contract and service-related customer groups.
3. To identify the customer groups with higher churn rates.
4. To build an interactive Power BI dashboard for analysis and presentation.
5. To suggest practical retention actions based on the patterns found.

3. Dataset Description

I used the Telco Customer Churn dataset [1] for this task. The dataset has 7,043 rows, where each row is one customer, and 21 columns. The columns cover customer information, the services each customer uses, contract details, billing information and whether the customer churned. A summary of the dataset is given in Table 1.

Table 1: Summary of the dataset

Item	Details
Dataset	Telco Customer Churn
Number of rows	7,043 (one row per customer)
Number of columns	21
Target column	Churn (Yes / No)
Main factors analyzed	Contract, Internet Service, Tenure, Monthly Charges, Payment Method, Tech Support, Online Security

Table 2 shows how the 21 columns can be grouped. Out of these, the analysis focused on seven factors: contract type, internet service, tenure, monthly charges, payment method, Tech Support and Online Security.

Table 2: Columns in the dataset, grouped by type

Group	Columns
Customer information	customerID, gender, SeniorCitizen, Partner, Dependents
Services	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Contract details	tenure (number of months with the company), Contract
Billing information	PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Churn status	Churn (target column)

4. Tools and Technologies Used

The following tools were used in this task:

- Python [2]: for inspecting, transforming, grouping and analyzing the data.
- Pandas [3]: for handling the data in table form and calculating the KPIs.
- Matplotlib [4]: for making exploratory charts during the analysis.
- Power BI Desktop [5]: for building the interactive dashboard and presenting the results.

5. Methodology

I completed the task in six main steps, which are described in Sections 5.1 to 5.6. The first four steps were carried out in Python, the fifth in Power BI Desktop, and the last one was about interpreting the results. The overall workflow was:

Load and inspect data → Prepare data → Calculate KPIs → Compare churn across groups → Build dashboard → Write insights and recommendations

5.1 Data Loading and Inspection

I loaded the dataset from the CSV file into Python using Pandas. I then checked its structure: the number of rows and columns, the data type of each column, and the values in the main categorical columns such as Contract, InternetService and PaymentMethod.

5.2 Data Preparation

The Churn column (Yes / No) was used as the main outcome variable throughout the analysis. Since tenure and monthly charges are numeric columns, I grouped them into ranges to create tenure groups and monthly-charge groups. This made it possible to compare churn between, for example, new customers and long-term customers.

5.3 KPI Calculation

I calculated four overall KPIs: total customers, churned customers, overall churn rate and average tenure. The churn rate was calculated using the formula below:

$$\text{Churn Rate (\%)} = (\text{Number of churned customers} / \text{Total number of customers}) \times 100$$

The same formula was used for each customer group, counting only the customers in that group. For example, the churn rate for month-to-month customers is the number of month-to-month customers who churned, divided by the total number of month-to-month customers.

5.4 Segment Analysis

Next, I compared churn rates across contract type, internet service, payment method, tenure groups, monthly-charge groups, Tech Support and Online Security. I also compared the average tenure of churned and retained customers. I used Matplotlib charts during this step to see the differences between groups visually.

5.5 Dashboard Development

After the analysis in Python, I built an interactive report in Power BI Desktop. It includes KPI cards, slicers for filtering, clustered column charts for comparing churn between groups, and a matrix that shows the churn rate for every combination of contract type and internet service. The dashboard is described in Section 8.

5.6 Business Interpretation

Finally, I reviewed the results to pick out the most important patterns and turned them into short insights and retention recommendations. These are covered in Sections 7 and 9.

6. Overall Churn KPIs

Table 3 shows the four overall KPIs for the whole dataset.

Table 3: Overall churn KPIs

KPI	Value	Meaning
Total Customers	7,043	Total number of customer records analyzed
Churned Customers	1,869	Customers with Churn = Yes
Churn Rate	26.54%	Percentage of customers who churned
Average Tenure	32.37 months	Average number of months customers have been with the company

Out of 7,043 customers, 1,869 churned and the remaining 5,174 stayed. This gives an overall churn rate of 26.54%, which means slightly more than one in every four customers left. The average tenure is 32.37 months, which is a little under three years. The overall churn rate is a useful reference point for the next section, since a group with a churn rate well above 26.54% can be considered a higher-risk group.

7. Analysis and Key Findings

In this section, I compare churn across different customer groups. Each churn rate here is the observed churn rate within that group, which means the percentage of customers in the group who churned. Table 4 brings the results together, and each factor is discussed below.

Table 4: Churn rate by customer group

Factor	Customer group	Churn rate
Contract type	Month-to-month	42.71%
	One-year	11.27%
	Two-year	2.83%
Internet service	Fiber optic	41.89%
	DSL	18.96%
	No internet service	7.40%
Tenure group	0-6 months	52.94%
	49-72 months	9.51%
Payment method	Electronic check	45.29%
	Bank transfer (automatic)	16.71%
Tech Support	Without Tech Support	41.64%
	With Tech Support	15.17%
Online Security	Without Online Security	41.77%
	With Online Security	14.61%

Note: These are observed churn rates. They describe patterns in this dataset and should not be taken as causes of churn without further analysis.

7.1 Contract Type

Churn drops sharply as the contract length increases. Month-to-month customers have a churn rate of 42.71%, compared with 11.27% for one-year contracts and only 2.83% for two-year contracts. This means month-to-month customers churn about 15 times as often as two-year customers. One possible reason is that month-to-month customers are not tied to a long contract, so it is easier for them to leave.

7.2 Internet Service

Fiber optic customers have a churn rate of 41.89%, which is more than double the rate for DSL customers (18.96%). Customers with no internet service have the lowest churn rate for this factor at 7.40%. The data does not show why fiber optic customers leave more often. It could be related to service quality, support or customer expectations, so I have suggested looking into the fiber optic experience in Section 9.

7.3 Customer Tenure

Tenure shows one of the clearest patterns. Customers with a tenure of 0-6 months have a churn rate of 52.94%, which means more than half of the newest customers churned. In comparison, only 9.51% of customers with a tenure of 49-72 months churned.

The average tenure comparison supports this. Churned customers had an average tenure of 17.98 months, while retained customers had an average tenure of 37.57 months, which is more than twice as long. Together, these results show that churn is concentrated among newer customers, and customers who have stayed longer are much less likely to leave.

7.4 Payment Method

Customers who pay by electronic check have a churn rate of 45.29%. This is much higher than the 16.71% churn rate of customers who pay by automatic bank transfer. The analysis does not explain the reason for this difference, but it does make electronic check users a group worth looking into.

7.5 Tech Support and Online Security

Both of these services show a similar pattern. Customers without Tech Support have a churn rate of 41.64%, compared with 15.17% for customers who have it. For Online Security, the churn rate is 41.77% for customers without the service and 14.61% for customers with it. In both cases, customers without the service churn more than 2.5 times as often as those who have it.

Here, “without” refers to customers who have internet service but did not take that service. Customers with no internet service at all are a separate group in the dataset, since these services need an internet connection.

7.6 Monthly Charges

I also compared churn across the monthly-charge groups created during data preparation (Section 5.2).

7.7 Contract Type and Internet Service Combined

The Power BI matrix shows the churn rate for each combination of contract type and internet service. This helps to find combinations where churn is especially high. For example, a month-to-month contract with fiber optic service brings together the highest-churn group from Section 7.1 and the highest-churn group from Section 7.2.

7.8 Summary of Findings

Looking at all the factors together, the groups with the highest churn are new customers (0-6 months), electronic check users, month-to-month customers, fiber optic users, and customers without Tech Support or Online Security. All of these groups have churn rates above 40%, which is well above the overall rate of 26.54%. On the other hand, customers on two-year contracts

(2.83%), customers with no internet service (7.40%) and long-term customers with 49-72 months of tenure (9.51%) have low churn rates.

8. Power BI Dashboard

I built the dashboard in Power BI Desktop. The report is divided into two pages so that the numbers and their interpretation are kept separate. The first page is for exploring and monitoring the data, while the second page explains the findings and recommendations. Table 5 lists what each page contains.

Table 5: Pages in the Power BI report

Page	Contents
Churn Dashboard	KPI cards; slicers for Contract Type, Internet Service and Customer Tenure; churn comparison charts; tenure and charge analysis; service-factor analysis; and the Contract x Internet Service churn-rate matrix.
Insights & Recommendations	Key findings and retention actions based on the observed patterns.

8.1 Page 1: Churn Dashboard

This page has KPI cards showing the overall figures from Section 6. The three slicers (Contract Type, Internet Service and Customer Tenure) are used to filter the visuals on the page. For example, selecting “Month-to-month” in the Contract Type slicer shows the figures for month-to-month customers only. The clustered column charts compare churn between groups, including the tenure and charge groups and the service factors, and the matrix shows the churn rate for each contract type and internet service combination.



Figure 1: Page 1 of the Power BI report (Churn Dashboard)

8.2 Page 2: Insights & Recommendations

The second page lists the key findings and the retention actions based on them. Keeping this on a separate page means the main conclusions can be read quickly without going through every chart.

Customer Churn Insights and Recommendations
KEY INSIGHTS • Overall churn rate is 26.54%, with 1,869 customers churned. • The 0–6 month tenure group has the highest observed churn rate at 52.94%. • Month-to-month customers show 42.71% observed churn, compared with 11.27% for one-year and 2.83% for two-year contracts. • Electronic check users show 45.29% observed churn. • Fiber optic customers show 41.89% observed churn. • Customers without Tech Support show 41.64% observed churn. • Customers without Online Security show 41.77% observed churn. • Churned customers have a shorter average tenure: 17.98 months versus 37.57 months for retained customers.
RETENTION RECOMMENDATIONS • Strengthen onboarding and regular check-ins during the first 6 months. • Provide suitable plan options and incentives for month-to-month customers considering longer-term contracts. • Review the electronic-check payment experience and promote convenient payment alternatives. • Investigate service quality and customer-support experience among fiber optic customers. • Increase awareness and availability of Tech Support and Online Security services. • Review pricing and perceived value for customers in the \$70–\$100 monthly-charge range. • Monitor high-risk combinations such as month-to-month customers using fiber optic service.

Figure 2: Page 2 of the Power BI report (Insights & Recommendations)

9. Retention Recommendations

Based on the findings in Section 7, I suggest the following actions to help reduce churn:

1. **Focus on new customers.** Churn is highest in the first six months (52.94%). Stronger onboarding and regular check-ins during this period could help more new customers stay.
2. **Improve retention of month-to-month customers.** These customers churn at 42.71%, much more than customers on longer contracts. The company could review the value of these plans, improve communication with these customers and offer clear upgrade options to one-year or two-year contracts.
3. **Review the electronic check payment experience.** Electronic check users churn at 45.29%. It is worth checking whether customers face any difficulty with this payment method, and making convenient alternative payment methods clearly available.
4. **Look into the fiber optic service experience.** Fiber optic customers churn at 41.89%, more than double the DSL rate. Service quality, support interactions and customer expectations for these customers should be examined.
5. **Promote Tech Support and Online Security.** Customers without these services churn at over 41%, compared with about 15% for those who have them. Making these services better known and easier to add for eligible customers may help.

6. **Review customers with high monthly charges.** The perceived value and service experience of customers in the higher monthly-charge groups should be checked (see Section 7.6).
7. **Monitor high-risk combinations.** Combinations such as a month-to-month contract with fiber optic service can be tracked using the dashboard matrix, so that these customers can be targeted for follow-up.

These recommendations are based on patterns observed in the data, so they should be treated as areas to investigate and test, not as guaranteed solutions.

10. Challenges Faced

During the completion of this project, I faced a few challenges while working with Python and Power BI:

- **Learning Power BI for the first time:** Understanding visuals, filters, slicers, formatting options, and the new Power BI interface required some practice.
- **Sorting categorical groups:** Arranging the tenure groups and monthly-charge groups in a logical ascending order required creating additional sorting fields.
- **Card number formatting:** Power BI initially displayed customer counts in abbreviated formats such as 7K and 2K, which required additional formatting and measures.
- **Measure vs. column confusion:** While creating calculated fields for the dashboard, distinguishing between measures and calculated columns was initially challenging.
- **Dashboard layout:** Fitting multiple KPIs, slicers, charts, and the heatmap into a clean and readable dashboard required adjusting the canvas size and visual alignment.
- **Interactive filtering:** Testing slicers and ensuring that the KPIs and charts updated correctly helped in understanding Power BI's interactive functionality.

11. Learning Outcomes

Through this task, I learned and practiced the following:

- Inspecting a real dataset in Python and calculating KPIs such as churn rate and average tenure using Pandas.
- Grouping numeric columns like tenure and monthly charges into ranges to compare churn between groups.
- Building an interactive dashboard in Power BI Desktop with KPI cards, slicers, clustered column charts and a matrix.
- Understanding the difference between a pattern in the data and a proven cause, and writing the results carefully because of this.
- Turning analysis results into practical recommendations for a business.

12. Limitations and Future Scope

This analysis has some limitations:

- It is a descriptive analysis based only on the patterns observed in this dataset.
- A higher churn rate in a group does not prove that a single service, payment method or contract type causes churn.
- Some factors overlap. For example, one customer can be new, on a month-to-month contract and using fiber optic internet at the same time. Simple group comparisons cannot separate the effect of each factor.

As future work, a predictive model could be built to test the combined effect of these factors and to estimate the churn risk of each customer more formally.

13. Conclusion

In this task, I completed a full workflow, starting from inspecting the customer churn data in Python and ending with an interactive dashboard in Power BI. The overall churn rate is 26.54%, but churn is not spread evenly across customers. The analysis showed strong differences in churn across contract type, tenure, internet service, payment method and the support and security services. Churn is much higher for new customers and month-to-month customers, and much lower for customers with long contracts or long tenure.

The dashboard makes it possible to explore these patterns interactively, and its second page turns the findings into practical areas for retention-focused investigation. Overall, this task gave me hands-on experience of using data analysis to understand a real business problem like customer churn.

14. References

- [1] Telco Customer Churn dataset. <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
- [2] Python Software Foundation, Python programming language. Available at: <https://www.python.org/>
- [3] The pandas development team, pandas documentation. Available at: <https://pandas.pydata.org/docs/>
- [4] The Matplotlib development team, Matplotlib: Visualization with Python. Available at: <https://matplotlib.org/>
- [5] Microsoft, Power BI Desktop. Available at: <https://powerbi.microsoft.com/>

Appendix: Project Files

The project files are organized as shown below.

Github URL: https://github.com/amrutha87084-tech/FUTURE_DS_02

```
FUTURE_DS_02/  
├── data/  
│   └── Telco-Customer-Churn.csv  
├── src/  
│   └── analysis.py  
├── results/  
│   └── FUTURE_DS_02_Customer_Churn_Analysis_Report.pdf  
├── FUTURE_DS_02_Churn_Analysis.pbix  
└── README.md
```